KAPIL BHATTARAI

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OBJECTIVE

Agricultural Data Scientist specializing in crop simulation modeling (DSSAT), geospatial analytics, and machine learning. Seeking opportunities to apply data-driven approaches for improved agricultural decision-making, precision nutrient management, and climate-smart farming systems.

EDUCATION

Master of Science in Agricultural & Biological Engineering *Jan 2025 – Present | GPA: 4.0/4.0*

University of Florida, Gainesville, FL — UF Fellowship Recipient

- Thesis research focused on DSSAT crop simulation modeling for agricultural decision-making.
- Contributing to the development of Graphical Visualization GB2 for DSSAT v4.8.6.

Bachelor of Science in Agriculture *2017 – 2023*

Agriculture and Forestry University (AFU), Rampur, Chitwan, Nepal

RESEARCH EXPERIENCE

Graduate Research Assistant *Jan 2025 – Present*

Department of Agricultural & Biological Engineering, University of Florida

- Conducting thesis research on DSSAT crop simulation modeling and its applications for agricultural decision-making under varying management and climate scenarios.
- Contributing to DSSAT v4.8.6 by developing an enhanced graphical visualization tool (GB2) with modern UI and improved data processing capabilities.
- Expanding technical capabilities in crop modeling using Fortran and Python for model development and data analysis.

Undergraduate Researcher *Feb 2022 – Jul 2022*

Prime Minister Agriculture Modernization Project (PMAMP), Chitwan, Nepal

- Completed research on "Precision Nitrogen Management in Spring Rice Using Decision Support Tools in Chitwan, Nepal."
- Identified optimized nitrogen application strategies that reduced input costs by an estimated 15–20% while maintaining yield.
- Performed mobile soil sampling, digital soil testing, and soil health campaigns across multiple sites.

PROFESSIONAL EXPERIENCE

Nursery Manager *Mar 2024 – Sep 2024*

Plant Paradise — Kadaghari, Kathmandu, Nepal

- Conducted nursery operations including plant propagation, production planning, and sales management.
- Managed website updates, social media operations, and digital marketing efforts to increase visibility.
- Provided leadership in human resource management, including staff training and development.
- Collaborated with the Director to ensure effective financial planning and budgeting.

Agriculture Intern *Aug 2022 – Aug 2023*

Ragolski & Son's Ltd — Netiv HaAsara, Israel

- Gained advanced agricultural knowledge and practical training in a leading Israeli agribusiness specializing in hybrid vegetable varieties and high-tech greenhouse operations.
- Specialized in production, nursery management, and marketing of cut flowers in controlled environments.
- Practiced coco-peat farming and precision growing systems at commercial scale.

Agriculture Intern *Feb 2022 – Jul 2022*

Prime Minister Agriculture Modernization Project (PMAMP) — Chitwan, Nepal

- Conducted training on mechanization of rice farming and nursery management for local farmers.
- Delivered training on off-season vegetable production, vermicomposting, and climate-smart agriculture practices.

PUBLICATIONS

Bhattarai, K., Pandey, K.R., Marahatta, S., Dhakal, M., Bhusal, N.R., & Thapa, R. (2024). *Precision Nitrogen Management in Spring Rice (Oryza sativa L.) Using Decision Support Tools in Chitwan, Nepal.* Turkish Journal of Agriculture-Food Science and Technology, 12(6), 955–965. Cited: 3.

Bhusal, N.R., Pokhrel, M.R., Chaudhary, R., **Bhattarai, K.**, & Timalsina, P. (2025). *Impact of Weed Management Practices on the Growth and Yield of Spring Rice in Pyuthan, Nepal.* Journal of Agricultural Sciences and Engineering, 7(3), 150–164. Cited: 1.

Regmi, B., **Bhattarai, K.**, Bhusal, N.R., Poudel, S.R., & Kunwar, S. (2022). *Effects of Various Fertilizer Combinations on Okra (Abelmoschus esculentus L.) Varieties for Dietary Fiber Content and Vegetative Characteristic.* Himalayan Journal of Science and Technology, 6(1), 75–86. Cited: 1.

Gyawali, K., Dhital, P.R., **Bhattarai, K.**, & Baral, S. *Effects of Ethephon and Pruning Practices on Sex Expression and Yield of Cucumber (Cucumis sativus L.) in Rupandehi, Nepal.* Turkish Journal of Agriculture-Food Science and Technology, 13(2), 346–352.

PRESENTATIONS & EVENTS

Poster: "Evaluating the Performance of the DSSAT Cropping System Model" *Mar 2025*
University of Florida, Gainesville, FL

Crop Physiology Training (1.5 ECTS) *Nov 2025*
Wageningen University PE&RC & UF ABE Department — Joint Program

23rd DSSAT Development Sprint (Virtual Participant) *Aug 2025*
University of Kentucky

DSSAT 2025 International Training Program *May 2025*
University of Georgia, Griffin, GA — 72 participants from 28 countries

HiPerGator Supercomputer Facility Visit *Mar 2025*
University of Florida

22nd DSSAT Development Sprint (In-Person Participant) *Jan 2025*
University of Florida

R Data Analysis Workshop — Trainer *2022*
Agriculture and Forestry University, Rampur, Chitwan — Organized by GAVS

SELECTED PROJECTS

DSSAT Graphical Visualization GB2 *2025 – Present*

- Building enhanced graphical visualization capabilities for DSSAT v4.8.6 using Python, serving the global crop modeling research community.

Rogcheck — Crop Disease Identification App

- Developed an Android app using CNN and TensorFlow Lite for offline crop disease detection across potato, pepper bell, and tomato — classifying 10 diseases.

DNA Sequencing Classifier

- Built a machine learning classifier for DNA sequencing data achieving 99.3% accuracy with confusion matrix evaluation across 7 classes.

TECHNICAL SKILLS

Crop Modeling:	DSSAT v4.8, Model Calibration & Validation, Scenario Analysis, Sensitivity Analysis, Field Trial Design
Programming:	Python, R/RStudio, SQL, Fortran (learning), Git/GitHub, Shiny Apps
Data Science & ML:	Scikit-learn, TensorFlow/TFLite, Predictive Modeling, Time Series Analysis, Data Visualization
Geospatial:	QGIS, ArcGIS, GeoPandas, Google Earth Engine, Remote Sensing, NDVI/Satellite Imagery, Spatial Statistics

CERTIFICATIONS

DSSAT 2025 International Training	University of Georgia, Griffin, GA (May 2025)
AI Fluency: Framework & Foundations	Anthropic
AI Fluency for Students	Anthropic
Basic Image Classification with TensorFlow	Coursera Project Network (Apr 2022)
Remote Sensing Image Acquisition, Analysis and Applications	University of South Wales (Mar 2022)
Data Analysis with R Programming	Coursera (Jun 2021)
Introduction to Machine Learning	Duke University (Apr 2021)
Working with Geospatial Data in Python	DataCamp (Apr 2021)
Getting Started with AWS Machine Learning	AWS (May 2021)

ACADEMIC PROFILES

Google Scholar | ORCID | ResearchGate | LinkedIn | GitHub